

两个凸集的和还是凸集: 设 $S_1 \subseteq \mathbb{R}^n, S_2 \subseteq \mathbb{R}^n$ 是两个凸集. 定义
定理 $S_1 + S_2 = \{x_1 + x_2 \mid x_1 \in S_1, x_2 \in S_2\}$ 则 $S_1 + S_2$ 也是凸集

证明: 任取 $p, q \in (S_1 + S_2)$ 则存在 $p_1 \in S_1, p_2 \in S_2$ 使 $p = p_1 + p_2$

且存在 $q_1 \in S_1, q_2 \in S_2$ 使 $q = q_1 + q_2$. 任取 $0 \leq \theta \leq 1$

$$\theta p + (1-\theta)q = \theta(p_1 + p_2) + (1-\theta)(q_1 + q_2) = (\theta p_1 + (1-\theta)q_1) + (\theta p_2 + (1-\theta)q_2)$$

故 $\theta p + (1-\theta)q \in (S_1 + S_2)$ 即 $S_1 + S_2$ 为凸集.

定义: 透视函数: $f: (\mathbb{R}^n \times \mathbb{R}_{++}) \rightarrow \mathbb{R}^n: f(x) = \frac{1}{x_{n+1}} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ 要求 $x_{n+1} \neq 0$

定理: 透视函数的像保凸性

设 $C \subseteq (\mathbb{R}^n \times \mathbb{R}_{++})$ 是凸集. 证明 $f(C)$ 也是凸集.

证明: 任取 $p, q \in f(C)$. 即存在 $p', q' \in C$ 使 $f(p') = p, f(q') = q$

任取 $0 \leq \theta \leq 1$, $\theta p + (1-\theta)q$ 是否位于集合 $f(C)$ 中

$$\text{设 } \alpha = \frac{\theta q'_{n+1}}{\theta q'_{n+1} + (1-\theta)p'_{n+1}} \quad \text{则 } f(\alpha p' + (1-\alpha)q') = \frac{\alpha \begin{bmatrix} p'_1 \\ \vdots \\ p'_n \end{bmatrix} + (1-\alpha) \begin{bmatrix} q'_1 \\ \vdots \\ q'_n \end{bmatrix}}{\alpha p'_{n+1} + (1-\alpha)q'_{n+1}}$$

$$= \frac{\alpha p'_{n+1}}{\alpha p'_{n+1} + (1-\alpha)q'_{n+1}} \cdot \frac{\begin{bmatrix} p'_1 \\ \vdots \\ p'_n \end{bmatrix}}{p'_{n+1}} + \frac{(1-\alpha)q'_{n+1}}{\alpha p'_{n+1} + (1-\alpha)q'_{n+1}} \cdot \frac{\begin{bmatrix} q'_1 \\ \vdots \\ q'_n \end{bmatrix}}{q'_{n+1}}$$

$$= \frac{\alpha p'_{n+1}}{\alpha p'_{n+1} + (1-\alpha)q'_{n+1}} \cdot p + \frac{(1-\alpha)q'_{n+1}}{\alpha p'_{n+1} + (1-\alpha)q'_{n+1}} \cdot q = \theta p + (1-\theta)q$$

$$\text{其中 } \frac{\alpha p'_{n+1}}{\alpha p'_{n+1} + (1-\alpha)q'_{n+1}} = \frac{\frac{\theta q'_{n+1}}{\theta q'_{n+1} + (1-\theta)p'_{n+1}} \cdot p'_{n+1}}{\frac{\theta q'_{n+1}}{\theta q'_{n+1} + (1-\theta)p'_{n+1}} \cdot p'_{n+1} + \frac{(1-\theta)p'_{n+1}}{\theta q'_{n+1} + (1-\theta)p'_{n+1}} \cdot q'_{n+1}}$$

$$= \frac{\theta}{\theta + (1-\theta)} = \theta$$

证明的灵感来自于小孔成像的示意图.